

V2V Communication-Assisted Federated Learning in Vehicular Networks

Chang Kyung Kim^{ID}, Kyungsoo Kim^{ID}, Junho Jeong^{ID}, TaeYoung Kim^{ID}, Shinyoung Cho^{ID}, and SuKyoung Lee^{ID}

Abstract—Federated learning (FL) enables vehicles to train machine learning (ML) models in a distributed manner for intelligent transportation system (ITS) services without sharing raw data, thereby preserving privacy and reducing communication overhead on roadside units (RSUs). However, due to their mobility and limited communication resources, vehicles participating in FL may be unable to transmit their updated models after local training if they move out of the RSU's communication range. This limitation not only increases the convergence time of the global model but also wastes the computing resources of the participating vehicles used for training, as their trained models cannot be used to update the global model. To address these issues, this paper proposes a vehicle-to-vehicle communication-assisted FL (V2V-FL) mechanism for vehicular networks, in which vehicles moving out of the RSU's coverage transmit their updated models to the RSU via a V2V link with limited communication resources, ensuring that more participating vehicles contribute their updated models to FL. An optimization problem for updated model forwarding with resource allocation is formulated to maximize the completion probability of participating vehicles while minimizing the total delay incurred during the training process. Since the formulated problem is a mixed-integer non-convex problem, it is decomposed into two subproblems, which are solved using the Karush-Kuhn-Tucker (KKT) conditions and a heuristic algorithm. Simulation results show that the proposed scheme outperforms other benchmark methods in terms of the total delay and convergence performance.

Index Terms—Federated learning, vehicular network, V2V communication, mobility, total delay.

I. INTRODUCTION

THE growth of artificial intelligence (AI) technologies leads to the advancement of intelligent transportation system (ITS) applications, including smart cities and connected autonomous vehicles. Machine learning (ML) plays a crucial role in ITS by enabling smart services such as object detection, traffic prediction, traffic sign classification, and path optimization [1], [2], [3]. However, the need for large amounts of

training data on ML models raises concerns regarding privacy and communication overhead in the vehicular networks [3], [4], [5], [6], [7].

Hence, federated learning (FL), which is a decentralized approach that trains local ML models on individual devices and aggregates them into a global model without sharing raw data, has emerged as a solution [8], [9]. FL is particularly important in vehicular networks because vehicles continuously generate large volumes of privacy and latency sensitive data (*e.g.*, driving behavior and camera images) and require periodic ML model updates to improve the user experience in intelligent transportation services [1], [4].

To reduce communication delays and improve the convergence rate of the training process in FL, existing studies have investigated client selection strategies for selecting participating devices with high communication resources, computational capacity, or data quality [2], [10]. However, unlike FL in stationary scenarios, vehicular networks are characterized by the dynamic nature of wireless communication due to vehicle mobility, which poses challenges in optimizing the communication delay and convergence rate during the FL training process [1], [8].

Therefore, several studies on client selection in vehicular networks have considered vehicle mobility when selecting optimal vehicles with adequate computing power, reliable communication resources, or sufficient data for the training process, including local training and global aggregation [11], [12], [13], [14], [15]. Zhang et al. [11] proposed a client selection method that considers the resource heterogeneity of clients in an ITS to improve the accuracy and convergence rate of a global model. Zhang et al. [12] proposed a vehicle selection scheme that enabled a roadside unit (RSU) to select suitable vehicles, minimizing the global loss function of the FL while ensuring a specific training delay threshold. Cha et al. [13] studied a vehicle selection method that considers not only the differences in data, communication resources, and computing capabilities of vehicles but also the freshness of the data collected by each vehicle. Xie et al. [14] analyzed the probability distribution of the number of vehicles that successfully upload their local models in each round. Fu et al. [15] studied a selective aggregation method that selects vehicles before uploading local models based on the performance of the trained local models, thereby reducing unnecessary communication delays.

However, due to the limited communication range of RSUs and the mobility of vehicles, some vehicles participating in the

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training process may move out of the communication range of the RSU. Consequently, they cannot transmit the updated model (*i.e.*, weight parameters) directly to the RSU, thereby failing to contribute to the training process, which degrades the accuracy and convergence rate of the model [10], [16]. In addition, if a vehicle is unable to transmit its training update to the RSU, the time spent on local training is wasted. Therefore, it is crucial to analyze the mobility and available communication resources of vehicles to ensure that they can transmit the updated model to the RSU after local training.

This motivates us to design a vehicle-to-vehicle communication-assisted FL (V2V-FL) mechanism that allows vehicles moving out of the RSU's communication range to forward their training updates to other vehicles before leaving the coverage area, thereby improving the accuracy and convergence performance of the model. We model the completion probability of a vehicle, which represents the likelihood of successfully completing local training and forwarding the updated model to the RSU with the assistance of multi-hop V2V communication in FL. We then formulate the problem of updated model forwarding and resource allocation to maximize the completion probability of vehicles while minimizing the total delay, which represents the time required to complete the FL training process. Finally, we solve the proposed problem using a combination of the Karush-Kuhn-Tucker (KKT) conditions and a heuristic algorithm. The main contributions of this study are summarized as follows:

- We propose a V2V-FL mechanism for vehicular networks to increase vehicle participation in FL by allowing updated models to be transmitted either directly to the RSU or via V2V communication. Considering the mobility of each vehicle, we develop the completion probability, which represents the likelihood of successful local training and model transmission, assisted by V2V communication. We then design a weighted averaging policy for FL model aggregation based on the completion probability of each vehicle to improve the accuracy of the global model.
- We formulate a model forwarding and resource allocation problem to maximize the completion probability and minimize the total delay. Since the formulated problem is a mixed-integer non-convex problem, we decompose it into forwarding decision and resource allocation subproblems. To address these subproblems, we propose a resource allocation algorithm that applies KKT conditions to assign communication resources to vehicles for transmission. In addition, we propose a heuristic forwarding algorithm to determine which vehicles transmit the updated model via V2V communication and to identify the recipient vehicles.
- The performance of the proposed method is evaluated using SUMO and OMNeT++ on a city-area map of Seoul, Korea. The numerical results demonstrate that the proposed algorithm outperforms benchmark methods in terms of the total delay, convergence performance, and test accuracy.

The remainder of this paper is organized as follows: Section II provides a literature review, and Section III introduces

the use cases of FL in vehicular networks and presents the system model. Section IV details how the V2V-FL optimization problem is formulated, and Section V explains the proposed algorithm used to solve the problem. In Section VI, we discuss the performance evaluation and simulation results. Finally, Section VII concludes the paper.

II. RELATED WORK

A. FL in Edge Networks

Due to the communication overhead caused by the increasing volume of data generated by smart devices and growing concerns regarding data security, FL has been proposed as a promising solution for training the ML models used in various smart applications within edge networks [8], [9]. However, it remains challenging to minimize the total training delay while optimizing the model performance, particularly with heterogeneous devices that possess varying computing and communication resources [2], [8], [9].

To address this challenge, some studies have explored strategies to reduce the total delay and convergence time in the FL training process by selecting participating clients based on their computing capabilities and communication resources [2], [5], [6], [7], [17], [18]. Wang et al. [5] examined the time consumption of previous communication rounds to select devices without analyzing local data or models to minimize both the total delay and convergence time. Huang et al. [6] proposed a cluster-based client selection approach to group clients with similar characteristics, aiming to reduce the total delay and communication costs. Werbi et al. [7] studied bilateral client selection by considering the preferences of both the edge server and client devices in the selection process while also accounting for newly joining IoT devices to enhance model accuracy. Trindade and da Fonseca [17] designed a client selection algorithm that adaptively determines the number of clients at different intervals, allowing it to adjust to dynamic system conditions, thereby reducing the communication resource usage between the edge server and user devices. Chen et al. [18] formulated a delay minimization problem that optimizes client selection and training procedures in federated learning by considering the heterogeneity of the collected data and the time required for local training on each device. However, it is difficult to adapt the client selection strategies proposed in these studies to vehicular networks because vehicle mobility leads to frequent topology changes, intermittent connectivity, and unpredictable client availability.

B. Client Selection for FL in Vehicular Networks

Due to the dynamic changes in communication resources caused by vehicle mobility, unstable communication links between vehicles and RSUs increase the time required to aggregate the global model from the selected vehicles, thereby increasing the convergence time and reducing the model accuracy. To address this challenge, recent studies have focused on vehicle selection, which determines the vehicles to participate in the FL training process while considering their mobility and the dynamics of communication resources [12], [14], [19], [20], [21]. Pervej et al. [19] studied a vehicle selection

scheme that considers the sojourn times and dataset sizes of vehicles under the RSU to select a subset of vehicles while ensuring the successful reception of trained models that meet the target accuracy under strict delay constraints. Sez et al. [20] studied a scheme for predicting vehicle residence times at base stations based on their speeds and positions to select participants for FL, considering the predicted residence times. Xie et al. [14] analyzed the probability distribution of the vehicles that successfully uploaded their local parameters in each round of FL to select the optimal participating vehicles and reduce the convergence rate. Zhang et al. [12] designed a probabilistic model that expresses the likelihood of a vehicle completing local training and forwarding the model parameters to the RSU before moving out of its communication range. Ballo et al. [21] proposed a resource scheduling scheme that determines when each vehicle sends model parameters to the RSU based on mobility patterns and available communication resources.

Although selecting vehicles with sufficient computing capabilities and communication resources can reduce the total delay and convergence time while maintaining model accuracy, the communication resources of vehicles must also be optimized because of the limited availability and mobility of these vehicles [22], [23], [24], [25]. Wu et al. [22] studied bandwidth and transmission power allocation strategies for moving vehicles to reduce the model update transmission delay between multiple vehicles and the RSU. Ji et al. [23] adopted a dynamic graph structure to analyze the state information available to vehicles, enabling them to autonomously select communication resources. Dai et al. [24] studied a resource allocation strategy that considers the heterogeneity of communication resources between vehicles and RSUs, as well as mobility, to maximize the quality of experience for the users of vehicular services.

While these studies successfully reduced the delay and improved the convergence rate of FL, they relied on infrastructure resources such as RSUs. However, with the increasing number of smart devices and vehicles, infrastructure resources alone cannot guarantee scalability or low-latency access to vehicles in the ITS [12], [19], [26].

C. V2V-Assisted Data Computation and Transmission

Since the communication resources of infrastructure elements such as RSUs are limited in handling a large number of vehicles and their requests, V2V communication has been adopted for data transmission in ITS services [27], [28], [29], [30]. Yan et al. [27] utilized a V2V communication-assisted method for task offloading to address the disconnection problem between vehicles and edge servers caused by vehicle mobility, by predicting the connection time between vehicles. Wu et al. [28] highlighted the under utilization of vehicle computing resources, and proposed a V2V-based computational offloading scheme to alleviate the computation and communication load on edge servers. Wu et al. [29] investigated V2V-assisted cooperative data caching schemes, in which vehicles can retrieve cached data from nearby vehicles to overcome the limited communication range of RSUs. Tan et al. [30] suggested a data retrieval scheme, such as

driving guidance, where data exchange between vehicles and RSUs is carried out not only through vehicle-to-infrastructure (V2I) communication but also via V2V-assisted communication to enhance smart services in the ITS.

The aforementioned studies investigated V2V-assisted communication for exchanging cached data and offloading computation tasks to alleviate the communication and computational burden on RSUs. However, none of these studies investigated V2V-assisted communication for transmitting updated model parameters to RSUs in FL. Unlike offloading and caching, which involve a single transmission per task or data request, FL requires iterative exchanges of updated models, which are communication-intensive and affect the convergence of the global model, thereby influencing the quality of smart services for all vehicles [12], [14], [19], [20], [21], [22], [23], [24], [25].

Motivated by the studies discussed above, we propose a V2V-FL mechanism that considers the mobility and available communication resources of vehicles during the training process. This mechanism enables vehicles that cannot forward their model parameters directly to the RSU to send them to other vehicles, which then transmit the parameters to the RSU. Hence, vehicles can still contribute to the FL training process even if they move outside the communication range of the RSU, leading to improved model accuracy and faster convergence rates in the training process.

III. USE CASE AND SYSTEM MODEL

In this section, we introduce the motivating use case and provide a detailed description of the proposed system model.

A. Use Case: FL in Intelligent Transportation System Services

FL has emerged as a promising approach for vehicular networks to enhance the efficiency and security of ML-based ITS services in smart cities, such as traffic target recognition, driver behavior analysis, predictive traffic congestion management, and dynamic route planning [1], [8], [15], [31]. FL can be applied to traffic flow prediction services that forecast traffic volumes and conditions, thereby enabling the optimization of traffic management systems. In traffic flow prediction applications, a model based on a convolutional neural network (CNN) or graph convolutional network (GCN) is utilized to analyze the graph structure properties of traffic data, and is periodically updated through FL to adapt to the dynamic nature of traffic patterns [1], [32].

As illustrated in Fig. 1, the FL training process for updating a traffic flow prediction model begins when the participating vehicles receive the base CNN/GCN model from an RSU. Each vehicle then performs local training using its data and subsequently sends the updated model back to the RSU upon completing the local training. Once the RSU collects all the updated models from the vehicles, it aggregates them into an updated global model.

Nevertheless, some vehicles may not be able to transmit their updated models to the RSU if they move outside the RSU coverage area, which increases the convergence rate of the

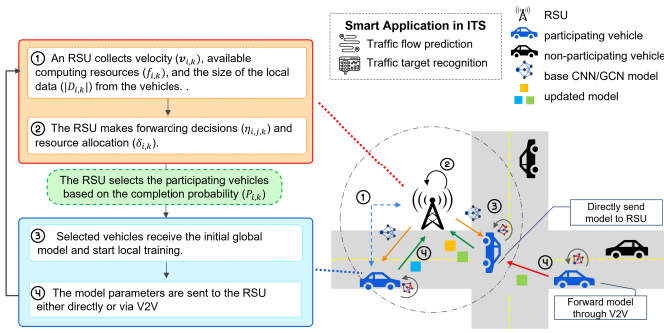


Fig. 1. Overall operation process of the V2V-FL mechanism in the vehicular network.

model and negatively affects the travel time and autonomous driving experience [12], [31]. These vehicles can forward the updated models to other vehicles within the communication range of the RSU and contribute to the training process, thereby enhancing the performance of the traffic flow prediction model, which adapts to the dynamic nature of traffic in a smart city.

B. System Description

We consider a vehicular network consisting of vehicles ($i \in I_v$) and an edge server located at an RSU deployed at the intersection of two-way roads in a smart city region where the maximum vehicle speed is 50 km/h [19], [33]. The RSU is responsible for determining the vehicles that participate in the FL and allocating communication resources to these selected vehicles. The RSU and vehicles use dedicated short-range communication (DSRC), based on IEEE 802.11p, for V2I and V2V communications [34], [35]. In DSRC for a vehicular network, the Wireless Access in Vehicular Environments (WAVE) Short Message Protocol (WSMP) defined by IEEE 1609.3 enables the exchange of short control and data messages between vehicles and RSUs [36]. The RSU can measure the received signal strength indicator (RSSI) values of vehicles by broadcasting WAVE short messages to nearby vehicles, and those with an RSSI above a predefined threshold are considered to be within its communication range.

As illustrated in Fig. 1, the FL training process is conducted periodically by the RSU through the following steps. (1) During each global iteration, the RSU initiates FL by broadcasting WAVE service advertisements to vehicles within its communication range, requesting their velocity ($v_{i,k}$), available computing resources (f_i), and training data size ($|D_{i,k}|$), where the WAVE message size is approximately 25–40 bytes (a payload of 12–24 bytes and an 8–12 bytes WSMP header) [35], [37]. (2) The RSU calculates the sojourn times of vehicles within its coverage area ($t_{i,k}^{\text{RSU}}$). (3) To maximize the completion probabilities of vehicles, the RSU determines the optimal forwarding decision and whether to send the updated model via V2I or V2V for each vehicle ($\eta_{i,j,k}$). It then allocates the optimal communication resource ($\delta_{i,k}$) based on their mobility and available computing and communication resources, which will be explained in more detail in Section V. (4) Using the determined resource allocation and forwarding

decision, the RSU calculates the completion probability ($P_{i,k}$), which represents the likelihood that a vehicle can contribute to FL, as described in Section IV-A. It then selects the vehicles whose completion probabilities exceed a predefined threshold. (5) The RSU sends the initial global model for the ITS services (e.g., a CNN/GCN model for traffic flow prediction) to the selected vehicles, along with forwarding and communication resource allocation decisions. Each selected vehicle then begins the local training using its own data. (6) After completing local training, the selected vehicles within the communication range of the RSU forward their updated models directly to the RSU. The model is assumed to be compressed or quantized before transmitting the model parameters to reduce their size, as in [38]. (7) Vehicles that leave the communication range of the RSU transmit the updated model to vehicle j via V2V communication, based on the forwarding decision received in Step 4. Specifically, they send route request messages to neighboring vehicles within communication range. Next, neighboring vehicles determine whether they can reach vehicle j directly or relay messages through neighboring vehicles. This process continues until the message reaches vehicle j within a maximum of H hops, at which point the updated model is sent to vehicle j . (8) The RSU aggregates the updated model received from the participating vehicles. The global aggregation process is repeated for K iterations, with newly selected vehicles at each iteration.

C. Computing and Transmission Model

We denote a set of an RSU and vehicles as $I = \{0, 1, 2, \dots, N\}$, where the RSU is represented by element 0, and the remaining elements 1 to N represent the vehicles. Let $I_v = \{i \mid i \in I, i \neq 0\}$ denote the subset of I that includes only the vehicles, excluding the RSU. At each global iteration $k \in K$, each vehicle $i \in I_v$ has a dataset $D_{i,k}$ with $|D_{i,k}|$ data points. Each data point contains a feature vector (e.g., historical traffic speed, vehicle density, and weather conditions) and a label that includes metadata (e.g., a timestamp and the location of the observed traffic flow).

Let f_i and α_i denote the CPU frequency of vehicle i and number of CPU cycles required to process one data sample in vehicle i , respectively. At each global iteration k , each vehicle i performs E stochastic gradient descent (SGD) iterations ($1 \leq e \leq E$) on its local dataset. Then, the computation time of vehicle i for executing local SGD iterations at each global iteration k is expressed as follows:

$$T_{i,k}^C = \frac{\alpha_i \cdot E \cdot |D_{i,k}|}{f_i}. \quad (1)$$

The data transmission rate of vehicle i depends on the communication resources allocated to it. We introduce continuous decision variable $\delta_{i,k} \in [0, 1]$, which represents the proportion of total communication resources allocated to vehicle i at global iteration k . Given these allocated resources, the transmission rate from vehicle i to entity j (either the RSU or another vehicle) is determined using Shannon's capacity formula:

$$r_{i,j,k} = \delta_{i,k} B \log_2 \left(1 + \frac{P_i g_{i,j}}{d_{i,j}^\gamma \sigma^2} \right), \quad \forall j \in I \setminus \{i\}, \quad (2)$$

TABLE I
LIST OF MAIN NOTATIONS

Symbol	Description
I, I_v, N	Set of RSU and vehicles; set of vehicles; total number of vehicles
K	Total number of global iterations
$D_{i,k}, D_k$	Dataset at vehicle i in iteration k ; total dataset in iteration k
f_i, α_i, E	CPU frequency of vehicle i ; required CPU cycles per sample; number of local SGD iterations
w_k	Global model weight at iteration k
$F_{i,k}(w), F_k(w)$	Local loss of vehicle i at iteration k ; global loss at iteration k
$\delta_{i,k}$	Fraction of total bandwidth allocated to vehicle i in iteration k
$p_i, g_{i,j}, d_{i,j}^\gamma$	Transmission power; average uplink gain; path loss
γ, B, σ	Path-loss exponent; system bandwidth; noise power
$r_{i,j,k}$	Transmission rate from vehicle i to entity j in iteration k
$t_{i,k}^{\text{RSU}}, t_{i,j,k}^{\text{V2V}}$	Sojourn time in RSU range; V2V sojourn time between i and j
G	Size of updated model parameters
$T_{i,k}^C, T_{i,k}^U$	Local computation time; uplink transmission time to the RSU
$T_{i,k}^F, T_{i,k}$	Reception time for updated model at vehicle i ; total delay for vehicle i
$\eta_{i,j,k}$	Indicator of forwarding updated model from i to j
$\mathbf{q}_{i,k}, \mathbf{v}_{i,k}$	Position and velocity of i at iteration k
$\mathbf{q}_{i,j,k}, \mathbf{v}_{i,j,k}$	Relative position and velocity between vehicle i and j at iteration k
R, R_i	Communication range of RSU; communication range of vehicle i
$R_{i,k}^h, H$	Set of vehicles reachable from i within h hops; number of maximum hops
$P_{i,k}^C$	Probability that vehicle i completes training and transmits to the RSU within its sojourn time
$P_{i,j,k}^F$	Probability of forwarding the updated model parameters to vehicle j within the sojourn time
$P_{i,k}$	Combined completion probability
$\epsilon, \alpha_{i,k}$	Threshold for vehicle selection; selection indicator in iteration k

where p_i denotes the transmission power of vehicle i , $g_{i,j}$ represents the average uplink channel gain between vehicles i and entity j , $d_{i,j}^\gamma$ denotes the path loss between vehicle i and entity j , γ is the path loss exponent, σ is the noise power, and B is the total system bandwidth.

The transmission delay during the global iteration includes two parts: 1) the delay caused by forwarding the updated model between vehicles and 2) the delay caused by forwarding the updated model from the vehicles to the RSU. Let $\eta_{i,j,k} \in \{0, 1\}$ be a binary indicator representing the forwarding of an updated model from vehicle i to nearby vehicle $j \in I_v \setminus \{i\}$ (i.e., $\eta_{i,j,k} = 1$ if vehicle i forwards the updated model to vehicle j at iteration k and $\eta_{i,j,k} = 0$ otherwise). Forwarding from vehicle i to vehicle j can occur in at most H hops through the intermediate vehicles between them. According to the IEEE 802.11p standard, an access delay occurs in DSRC communication to establish a channel connection between vehicles or between a vehicle and an RSU. The average access delay, denoted by $\hat{t}_{\text{dsrc}}$, is modeled using a Pareto-type tail with exponent θ : $P(A > t) = (t_0/t)^\theta$, $t \geq t_0$, $\theta > 1$, where A denotes the random variable representing the access delay, t denotes the delay threshold for the exceedance probability of access delay, and t_0 denotes the minimum possible access delay [39].

Then, the transmission time for sending an updated model of size G from vehicle i to vehicle j in V2V communication is

$$T_{i,j,k}^F = \sum_{[i',j'] \in l(i,j)} \frac{G \cdot \eta_{i,j,k}}{r_{i',j',k}} + \hat{t}_{\text{dsrc}}, \quad \forall j \in I_v \setminus \{i\}, \quad (3)$$

where i' and j' are the intermediate vehicles along the shortest multi-hop path from vehicle i to vehicle j . The notation $[i', j']$ represents a single hop, indicating that vehicle i' directly forwards the data to vehicle j' . Then, set $l(i, j)$ comprises all of the hops that form the shortest multi-hop path from vehicle i to vehicle j , ensuring that $|l(i, j)| < H$.

Based on Eq. (3), the total time required to receive the updated models from a set of nearby vehicles to vehicle i at global iteration k is

$$T_{i,k}^F = \max_{j \in I_v \setminus \{i\}} \{T_{j,i,k}^F\}. \quad (4)$$

The transmission time from vehicle i to the RSU is given by:

$$T_{i,k}^U = \frac{\sum_{j \in I_v \setminus \{i\}} G \cdot \eta_{j,i,k} + G \cdot \prod_{j \in I_v \setminus \{i\}} (1 - \eta_{j,i,k})}{r_{i,0,k}} + \hat{t}_{\text{dsrc}}, \quad (5)$$

where $r_{i,0,k}$ is the transmission rate between the vehicle and RSU. The total delay, which is the time required for the local training of the model and forwarding of the updated model for vehicle i , is

$$T_{i,k} = T_{i,k}^C + T_{i,k}^F + T_{i,k}^U. \quad (6)$$

Since the total delay is considered only for the selected vehicles, the total delay taken for aggregation at each global iteration k is

$$T_k = \max_{i \in I_v} \{a_{i,k} \cdot T_{i,k}\}, \quad (7)$$

where $a_{i,k} \in \{0, 1\}$ is an indicator function such that $a_{i,k} = 1$ if vehicle i is selected at iteration k , and $a_{i,k} = 0$ otherwise.

D. Mobility Model

1) *Mobility in V2I Communication:* To represent vehicle mobility in a vehicular network, the coordinates of each vehicle $i \in I_v$ at the start of global iteration k are denoted by vector $\mathbf{q}_{i,k} = (x_{i,k}, y_{i,k})$, with the origin $(0, 0)$ located at the RSU. Since the RSU does not have information regarding the destination of the vehicle, it estimates the guaranteed sojourn time, denoted as $t_{i,k}^{\text{RSU}}$. The coverage area of the RSU is represented by the equation $x^2 + y^2 \leq R^2$, where x and y denote the coordinates of a point within the communication range of RSU, and R represents the communication range of the RSU. The shortest distance from vehicle i to the communication range of RSU can then be expressed as $R - \|\mathbf{q}_{i,k}\| = R - \sqrt{x_{i,k}^2 + y_{i,k}^2}$, where $\|\cdot\|$ is the Euclidean norm of the vector. Let $\mathbf{v}_{i,k} = (u_{i,k}, v_{i,k})$ denote the velocity vectors of vehicle i at global iteration k , where $u_{i,k}$ and $v_{i,k}$ represent the velocity components in the horizontal and vertical directions, respectively. Subsequently, the minimum expected sojourn time of each vehicle i in the communication range of the RSU at iteration k is

$$t_{i,k}^{\text{RSU}} = \frac{R - \|\mathbf{q}_{i,k}\|}{\|\mathbf{v}_{i,k}\|}, \quad (8)$$

Since vehicle i must complete its local training and forward the updated model to the RSU while remaining within communication range of the RSU, the following constraint is applied:

$$T_{i,k} \leq t_{i,k}^{\text{RSU}}. \quad (9)$$

2) *Mobility in V2V Communication*: Considering the trajectory at the intersection of a smart city road, the relative location vector $\mathbf{q}_{i,j,k}$ and velocity vector $\mathbf{v}_{i,j,k}$ of vehicle $j \in I_v \setminus \{i\}$, relative to vehicle $i \in I_v$, at the start of global iteration k , are defined as $\mathbf{q}_{i,j,k} = \mathbf{q}_{j,k} - \mathbf{q}_{i,k}$ and $\mathbf{v}_{i,j,k} = \mathbf{v}_{j,k} - \mathbf{v}_{i,k}$, respectively. Then, the minimum expected sojourn time $t_{i,j,k}^{\text{V2V}}$ for V2V communication between vehicles i and j at iteration k is given by:

$$t_{i,j,k}^{\text{V2V}} = \frac{-\mathbf{q}_{i,j,k} \cdot \mathbf{v}_{i,j,k} + \sqrt{\|\mathbf{v}_{i,j,k}\|^2 R_i^2 - \|\mathbf{q}_{i,j,k} \times \mathbf{v}_{i,j,k}\|^2}}{\|\mathbf{v}_{i,j,k}\|^2}, \quad (10)$$

where $\mathbf{q}_{i,j,k} \cdot \mathbf{v}_{i,j,k}$ represents the rate of change of the relative distance between the vehicles, indicating whether the vehicles are moving closer together or farther apart, and $\mathbf{q}_{i,j,k} \times \mathbf{v}_{i,j,k}$ is the perpendicular component of their relative motion, which influences the duration for which vehicle j remains within the communication range, R_i , of vehicle i .

As the updated model forwarding via V2V communication must be completed when two vehicles are in communication range of each other, we have the following constraint:

$$T_{i,k}^F \leq t_{i,j,k}^{\text{V2V}}, \quad \forall j \in I_v \setminus \{i\}. \quad (11)$$

E. FL Model

The model training begins with the RSU broadcasting the initial global model to the participating vehicles. Each vehicle then begins local training using the received model, which has a model dimension of M and is associated with a loss function, $f(w, d)$, that quantifies the error of parameter $w \in \mathbb{R}^M$.

The local loss at vehicle i during iteration k is denoted as $F_{i,k}(w) \triangleq F_i(w | D_{i,k})$, where

$$F_i(w | D_{i,k}) = \frac{1}{|D_{i,k}|} \sum_{d \in D_{i,k}} f(w, d). \quad (12)$$

Subsequently, the global loss of the ML model at iteration k is defined as follows:

$$F_k(w) = \frac{1}{|D_k|} \sum_{i \in I_v} |D_{i,k}| F_{i,k}(w), \quad (13)$$

where D_k is the set of data in iteration k , i.e., $D_k = \sum_{i \in I_v} D_{i,k}$. The goal of the training process in FL is to find the optimal global weight parameters, which are given by

$$w_k^* = \arg \min_{w \in \mathbb{R}^M} F_k(w), \quad \forall k. \quad (14)$$

To simplify the theoretical analysis of the loss function, we assume that loss functions in all iterations are L -smooth and μ -strongly convex [12], [16].

At each global iteration k , each vehicle i performs E SGD iterations on the local dataset. Let $B_{i,j}^e$ be the data batch sampled in the e -th iteration. The evolution of the local parameters can be expressed as

$$w_{i,k}^{e+1} = w_{i,k}^e - l \cdot \nabla F_i(w_{i,k}^e | B_{i,j}^e), \quad (15)$$

where l denotes the learning rate. Therefore, the local weight parameters of vehicle i after E iterations are given by $w_{i,k} = w_{i,k}^E$.

IV. VEHICLE SELECTION AND PROBLEM FORMULATION

This section introduces the completion probability of a vehicle, which represents the likelihood that a vehicle can complete local training and forward its updated model to the RSU within a maximum of H hops in multi-hop V2V communication. Then, the vehicle selection method is explained based on the completion probability.

A. Vehicle Selection Scheme

To select vehicles expected to complete local training and forward updated models to the RSU via multi-hop V2V communication under constraints Eqs. (9) and (11), we design a completion probability model. The number of vehicles passing through the RSU per unit time follows a Poisson process [14], [40].

Assuming that a channel gain $g_{i,j}$ follows a Rayleigh distribution, the signal-to-noise ratio (SNR) is exponentially distributed. Hence, the probability that the vehicle can complete local training and forward the updated model to the RSU while it is within the communication range of the RSU is given by the cumulative distribution function (CDF) for the exponential distribution:

$$\begin{aligned} P_{i,k}^C &= P(T_{i,k} < t_{i,k}^{\text{RSU}}) \\ &= 1 - \exp(-\lambda_1 (t_{i,k}^{\text{RSU}} - T_{i,k})). \end{aligned} \quad (16)$$

where λ_1 represents the vehicle arrival rate at the RSU.

Similarly, the probability that the transmission time for forwarding via V2V communication is less than the sojourn time of the corresponding vehicle is given by

$$\begin{aligned} P_{i,j,k}^F &= P(T_{i,j,k}^F < t_{i,j,k}^{\text{V2V}}) \\ &= 1 - \exp(-\lambda_2 (t_{i,j,k}^{\text{V2V}} - T_{i,j,k}^F)). \end{aligned} \quad (17)$$

where $\lambda_2 = \frac{\lambda_1 \pi R_i^2}{vb}$ represents the rate of vehicles within the V2V communication range, v denotes the average vehicle speed in the area, and b is the width of the road. Subsequently, to consider multi-hop V2V communication, we extended the forwarding probability to account for up to H hops. Let $R_{i,k}^h$ be the set of vehicles reachable within h hops from vehicle i . The probability of successful forwarding from vehicle i to vehicle j across h hops, denoted as $P_{i,j,k}^{F,h}$, is defined recursively as follows:

$$P_{i,j,k}^{F,h} = 1 - \prod_{j' \in R_{i,k}^{h-1}} (1 - P_{j',j,k}^F), \quad \text{for } h > 1. \quad (18)$$

The combined probability $P_{i,k}$ represents the likelihood that vehicle i either completes its updated model within its sojourn time in the RSU, $t_{i,k}^{\text{RSU}}$, or successfully forwards the updated model, which is then completed through multi-hop V2V communication involving other vehicles within h hops.

The combined completion probability considering multi-hop V2V forwarding is given by

$$P_{i,k} = P_{i,k}^C + (1 - P_{i,k}^C) \sum_{h=1}^H \left(1 - \prod_{j \in R_{i,k}^h} (1 - P_{i,j,k}^{F,h}) \right). \quad (19)$$

Finally, we define an indicator function to determine the selection of vehicles in the global aggregation as

$$a_{i,k} = \mathbf{1}(P_{i,k} \geq \epsilon), \quad (20)$$

where $\mathbf{1}(\cdot)$ is an indicator function that takes the value of either 0 or 1, and ϵ is the probability threshold. Therefore, only vehicles with a completion probability that exceeds the threshold ϵ are selected to participate in FL training.

Let $I_{i,k}$ be the set of vehicles within the communication range of vehicle i at iteration k . Considering the completion probability $P_{i,k}$ during global aggregation, the weighted averaging policy for FL model aggregation can be expressed as

$$w_{k+1} = \sum_{i \in I_{i,k}} \left(\frac{|D_{i,k}|/P_{i,k}}{\sum_{j \in I_{i,k}} |D_{j,k}|/P_{j,k}} \cdot w_{i,k} \right). \quad (21)$$

B. Problem Formulation

In this paper, we aim to minimize the total delay, maximize the completion probability of vehicles, and ensure efficient convergence of the global model. The optimization problem is formulated as follows:

$$\mathcal{P} : \min_{\delta_{i,k}, \eta_{i,j,k}} \frac{1}{K} \sum_{k=0}^{K-1} \left(\omega T_k + (1 - \omega) \sum_{i \in I_v} \rho_{i,k} \frac{1}{P_{i,k}} \right) \quad (22)$$

$$\text{s.t. (9), (11),}$$

$$\sum_{i \in I_v} \delta_{i,k} = 1, \quad (23)$$

$$\eta_{i,j,k} \in \{0, 1\}, \quad \forall i, j, k, \quad (24)$$

where $\rho_{i,k} = \frac{|D_{i,k}|}{\sum_{j \in I_{i,k}} |D_{j,k}|}$ and ω is a weight factor. Eq. (23) ensures that the allocated communication resources are fully utilized while remaining within the available capacity of the system. Eq. (24) is a feasibility constraint on forwarding decision variable $\eta_{i,j,k}$. The problem $\mathcal{P}$ is difficult to solve because of the joint optimization of the resource allocation and forwarding decisions for the updated model under dynamic mobility and strict latency constraints, making it highly non-linear and complex. To solve this problem, we decompose it into two subproblems to optimize resource allocation $\delta_{i,k}$ and forwarding decision $\eta_{i,j,k}$.

V. RESOURCE ALLOCATION AND FORWARDING ALGORITHM

In this section, we decompose the problem $\mathcal{P}$ into two subproblems, as it is a mixed-integer non-convex problem. We introduce resource allocation and updated model forwarding decision algorithms that minimize the total delay and improve the completion probability by solving each sub-problem separately.

A. Resource Allocation Algorithm

We introduce the resource allocation subproblem $\mathcal{P}_1$ to optimize the communication resources for each vehicle i at iteration k , denoted by $\delta_{i,k}$, under the assumption that $\epsilon \geq \frac{1}{1+\alpha T_{i,k}}$ in Eq. (20), where α is a scaling parameter:

$$\mathcal{P}_1 : \min_{\delta_{i,k}} \frac{1}{K} \sum_{k=0}^{K-1} \left(\omega T_k + (1 - \omega) \sum_{i \in I_v} \rho_{i,k} (1 + \alpha T_{i,k}) \right) \quad (25)$$

s.t. (9), (11), (23).

Since the objective function is convex and all constraints form a convex feasible set, the subproblem $\mathcal{P}_1$ is convex, allowing the KKT conditions to be applied to obtain the optimal solution. To simplify $\mathcal{P}_1$ by isolating the resource allocation decision $\delta_{i,k}$, we first fix all $\eta_{i,j,k}$ to 1.

The Lagrangian function of subproblem $\mathcal{P}_1$ can be expressed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{K} \sum_{k=0}^{K-1} \left(\omega \cdot T_k + (1 - \omega) \cdot \sum_{i \in I_v} \rho_{i,k} (1 + \alpha T_{i,k}) \right) \\ & + \lambda_{i,j,k} (T_{i,j,k}^F - t_{i,j,k}^{V2V}) + \lambda_{i,k} (T_{i,k} - t_{i,k}^{\text{RSU}}) \\ & + \lambda_k \left(\sum_{i \in I_v} \delta_{i,k} - 1 \right), \end{aligned} \quad (26)$$

where $\lambda_{i,j,k}$, $\lambda_{i,k}$, and λ_k are the Lagrangian multipliers for the corresponding constraints. Then, the KKT conditions of the problem are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \delta_{i,k}} = & \frac{1}{K} \sum_{k=0}^{K-1} \left(\omega \cdot \frac{\partial T_k}{\partial \delta_{i,k}} + (1 - \omega) \cdot \sum_{i \in I_v} \rho_{i,k} \cdot \alpha \cdot \frac{\partial T_{i,k}}{\partial \delta_{i,k}} \right) \\ & + \lambda_{i,j,k} \frac{\partial T_{i,j,k}^F}{\partial \delta_{i,k}} + \lambda_{i,k} \frac{\partial T_{i,k}}{\partial \delta_{i,k}} + \lambda_k, \end{aligned} \quad (27)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_{i,j,k}} = T_{i,j,k}^F - t_{i,j,k}^{V2V}, \quad (28)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_{i,k}} = T_{i,k} - t_{i,k}^{\text{RSU}}, \quad (29)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_k} = \sum_{i \in I_v} \delta_{i,k} - 1. \quad (30)$$

We can obtain $\delta_{i,k}$ by solving the equations that Eqs. (27)-(30) are zeros for three feasible cases.

1) *Case 1:* $\lambda_{i,j,k} > 0$, $\lambda_{i,k} = 0$, and $\lambda_k > 0$: Only the V2V forwarding constraint (Eq. (28)) is active. Therefore, the resource allocation decision $\delta_{i,k}$ is made to satisfy the V2V transmission deadline as follows:

$$\delta_{i,k} = \frac{G}{t_{i,j,k}^{V2V} \cdot r_{i,j,k}}, \quad (31)$$

which gives

$$N \cdot t_{i,j,k}^{V2V} + T_{i,k}^C \leq t_{i,k}^{\text{RSU}}. \quad (32)$$

2) *Case 2:* $\lambda_{i,j,k} = 0$, $\lambda_{i,k} > 0$, and $\lambda_k > 0$: Only the RSU sojourn constraint (Eq. (29)) is active. Hence, the allocation is adjusted to ensure completion before the vehicle leaves RSU coverage as follows:

$$\delta_{i,k} = \frac{G(N-1)}{r_{i,j,k} \cdot (t_{i,k}^{\text{RSU}} - T_{i,k}^C)}, \quad (33)$$

Algorithm 1 Resource Allocation Algorithm

```

1: Input:  $K, G, r_{i,j,k}, r_{i,0,k}, t_{i,k}^{RSU}, I, t_{i,j,k}^{V2V}, \alpha_i, |D_{i,k}|, f_i$ 
2: Output: Optimal  $\delta_{i,k}^*$ 
3: for  $i \in I$  do
4:   Computes  $\lambda_{i,j,k}, \lambda_{i,k}, \lambda_k$  using bisection method.
5:   if Eq. (32) is satisfied then
6:      $\delta_{i,k} \leftarrow \frac{G}{t_{i,j,k}^{V2V} \cdot r_{i,j,k}}$ 
7:   else if Eq. (34) is satisfied then
8:      $\delta_{i,k} \leftarrow \frac{G(N-1)}{r_{i,j,k} \cdot (t_{i,k}^{RSU} - T_{i,k}^C)}$ 
9:   else
10:     $\delta_{i,k} \leftarrow \sqrt{\frac{G(N+2\rho_{i,k} \cdot \alpha \cdot (N-1))}{\lambda_k \cdot K \cdot r_{i,j,k}}}$ 
11:   end if
12: end for

```

which leads to

$$t_{i,j,k}^{V2V} \cdot (N-1) + T_{i,k}^C \geq t_{i,k}^{RSU}. \quad (34)$$

3) *Case 3:* $\lambda_{i,j,k} = 0$, $\lambda_{i,k} = 0$, and $\lambda_k > 0$: Neither the V2V forwarding nor the RSU sojourn constraints (Eqs. (28) and (29)) are binding. Therefore, the resource allocation is assigned by the overall resource availability as

$$\delta_{i,k} = \sqrt{\frac{G(N+2\rho_{i,k} \cdot \alpha \cdot (N-1))}{\lambda_k \cdot K \cdot r_{i,j,k}}}. \quad (35)$$

The proposed algorithm determines the optimal resource allocation decision, $\delta_{i,k}$, based on the KKT conditions, as presented in Algorithm 1.

B. Updated Model Forwarding Algorithm

After resource allocation $\delta_{i,k}^*$ is obtained from subproblem $\mathcal{P}_1$, we proceed to subproblem $\mathcal{P}_2$ to determine the optimal updated model forwarding strategy, $\eta_{i,j,k}$, as follows:

$$\begin{aligned} \mathcal{P}_2 : \min_{\eta_{i,j,k}} \quad & \frac{1}{K} \sum_{k=0}^{K-1} \left(\omega T_k + (1-\omega) \sum_{i \in I_v} \rho_{i,k} (1 + \alpha T_{i,k}) \right) \\ \text{s.t.} \quad & (9), (11), (24). \end{aligned} \quad (36)$$

$\mathcal{P}_2$ is a combinatorial optimization problem with binary decisions constrained by communication and mobility constraints, making it computationally complex [12]. To solve this problem, we propose a heuristic forwarding decision algorithm (Algorithm 2), which operates through the following steps. In each iteration k , after the RSU makes resource allocation decisions $\delta_{i,k}^*$ (line 3), it selects vehicle j' to which vehicle i sends the updated model via V2V communication. This selection is made by identifying vehicles with a higher completion probability, $P_{j,k}$, while also considering their proximity to vehicle i (line 7). The algorithm then checks whether destination vehicle j satisfies the constraints of Eqs. (9) and (11) (line 8). If feasible, it records forwarding decision $\eta_{i,j',k} = 1$ and stops searching for vehicle i (lines 9-10). If no suitable neighbor satisfies the V2V constraint, the algorithm removes j' from the candidate vehicle set I'_v (line 12). This process continues until either a suitable destination vehicle, j' , is found or no suitable neighbor is identified. In the latter case, vehicle i does

Algorithm 2 Forwarding Decision Algorithm

```

1: Input:  $I_v, \epsilon, t_{i,k}^{RSU}, t_{i,j,k}^{V2V}, T_{i,k}^U, T_{i,k}^C$ 
2: Output:  $\{a_{1,k}, \dots, a_{N,k}\}$ 
3: Set  $\delta_{i,k}^*$ ,  $\forall i \in I_v$  based on Algorithm 1
4: for  $i \in I_v$  do
5:   Initialize candidate set:  $I'_v \leftarrow \{j \mid j \in R_{i,k}^H \setminus \{i\}\}$ 
6:   while  $I'_v \neq \emptyset$  do
7:     Find  $j' = \arg \max_{j \in I'_v} \frac{P_{j,k}}{\|q_{i,j,k}\|}$ 
8:     if Eqs. (9) and (11) are satisfied for  $j'$  then
9:       Set  $\eta_{i,j',k} \leftarrow 1$ 
10:      break
11:    else
12:      Remove  $j'$  from  $I'_v$ 
13:    end if
14:  end while
15:  if  $I'_v = \emptyset$  then ▷ No suitable neighbor found
16:    Set  $\eta_{i,j,k} = 0$ ,  $\forall j \in I_v$ 
17:  end if
18:  # Select the participating vehicles
19:  Calculate  $P_{i,k}$ ,  $\forall i \in I_v$  using Eq. (19)
20:  Set  $a_{i,k}$ ,  $\forall i \in I_v$  using Eq. (20)
21: end for

```

not forward the updated model anywhere (lines 15-16). After making resource allocation decisions ($\delta_{i,k}$) and forwarding decisions ($\eta_{i,j,k}$), the RSU calculates completion probability $P_{i,k}$ for each vehicle i to select participating vehicles (line 19). Subsequently, it determines vehicle selection decision $a_{i,k}$ based on the calculated $P_{i,k}$ as in Eq. (20) (line 20). Finally, the vehicles selected in Algorithm 2 begin local training and transmit their updated models to the RSU, as shown in Fig. 1. The RSU then aggregates these models to update the global model using Eq. (21).

C. Complexity Analysis

In Algorithm 1, resource allocation is computed by evaluating the feasible case for each of the N vehicles, while the bisection method requires a finite number of iterations independent of N [41]. Therefore, the computational complexity of Algorithm 1 is $\mathcal{O}(N)$. For the forwarding decision in Algorithm 2, each of the N vehicles examines its candidate neighbors within H hops, where the maximum number of candidates is N , resulting in a time complexity of $\mathcal{O}(N^2)$. Finally, the overall complexity of the proposed framework combines both algorithms, resulting in $\mathcal{O}(N + N^2) = \mathcal{O}(N^2)$.

VI. PERFORMANCE EVALUATION

A. Simulation Setup

A simulation is conducted on the streets of Gangnam, covering a $5 \text{ km} \times 5 \text{ km}$ area near the Yeoksam subway station ($37.5^\circ, 127.0^\circ$) in Seoul, a district known for adopting smart vehicle technologies such as a late-night autonomous taxi service [43]. The road network is obtained from OpenStreetMap (OSM) and configured using the OSM Web Wizard to mimic the traffic layout and conditions of the Gangnam area, with two-lane roads in both directions, traffic signs, and

TABLE II
SIMULATION PARAMETERS [12], [19], [34], [42]

Parameter	Value
Total number of global iterations K	150-200
Maximum vehicle's velocity	50 km/h
CPU frequency f_i	0.5–5.5 GHz
RSU communication range R	1000 m
V2V communication range R_i	300 m
Size of model parameters G	1 Mbit
Maximum hops H	3
Transmission power p_i	1.3 W
System bandwidth B	200 MHz
Noise power σ	7.9×10^{-13} W
Path loss exponent γ	3.5
Batch size, Learning rate l	64, 0.1
Symmetric Dirichlet factor	0.1, 1
Access delay for DSRC t_{dsr}	1-20 ms

multiple intersections. The vehicle mobility is generated using SUMO's built-in traffic generation utility [44], ensuring spatial and temporal diversity. It follows the Intelligent Driver Model (IDM) with a maximum speed of 50 km/h and an average vehicle density of 50 vehicles/km², reflecting traffic conditions in Gangnam [45].

Vehicular communications for the model transmission between the vehicles and RSU are generated using OMNeT++ [46]. The CPU frequency f_i for each vehicle ranges from 0.5 to 5.5 GHz, with increments of 0.5 GHz. We set the RSU communication range to $R = 1000$ m, the vehicle transmission power to $p_i = 1.3$ W, and the system bandwidth to $B = 200$ MHz. The V2V communication range R_i and the maximum number of hops H are set to 300 m and 3, respectively [34], [42].

We evaluate MNIST, Fashion-MNIST (FMNIST), and the LISA Traffic Light dataset under both IID and non-IID splits, where MNIST and FMNIST serve as baselines [12] and LISA represents a vehicular task for comparison [47]. We use a CNN model with two convolutional layers, each followed by max-pooling layers, one fully connected layer, and an output layer. Both datasets are divided into 60,000 training and 100,000 testing samples. We set the batch size to 64 and the learning rate to 0.1. For the non-IID data, we employ a symmetric Dirichlet distribution to allocate labels across vehicles [19]. The remaining simulation parameters are listed in Table II.

We compare our proposed scheme with four benchmark methods: Random, Data-Aware (DA) [48], [49], Mobility and communication resource-aware (MACA) [12], and Even Resource Allocation (ERA) [19]. In the Random method, vehicles for local training are selected randomly, while in DA, vehicles are chosen based on the amount and distribution of the collected data to improve the accuracy of the global model. In MACA, only vehicles that satisfy the condition $P(T_{i,k} < t_{i,k}^{\text{RSU}})$ are selected, without utilizing V2V communication. The ERA method optimizes only the data forwarding decisions of vehicles, while the communication resources of each vehicle i are evenly distributed among all nearby vehicles and the RSU.

B. Simulation Results

1) *Evaluations on MNIST and FMNIST Datasets:* Fig. 2 illustrates the convergence performance of the loss value and

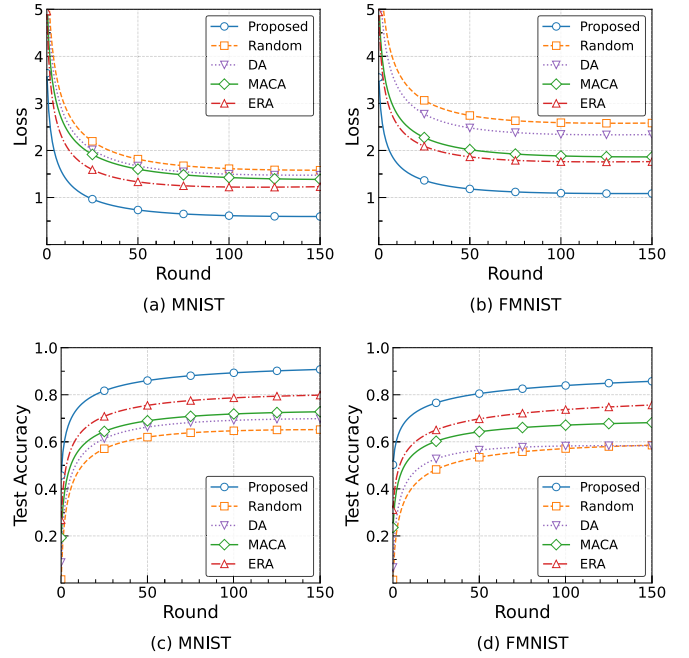


Fig. 2. FL performance comparison of different schemes over the MNIST and FMNIST datasets for IID.

test accuracy as the number of global rounds, K , increases with the IID of the MNIST and FMNIST datasets. We can observe that the proposed algorithm outperforms the three benchmark methods in terms of both the loss value and test accuracy in every round. Specifically, the proposed algorithm achieves higher average test accuracy than Random, DA, MACA, and ERA by 35.7%, 31.0%, 26.4%, and 16.3% on MNIST, respectively, and by 37.8%, 35.3%, 16.3%, and 25.3% on FMNIST, respectively. Random results in the highest loss value and lowest test accuracy because the participating vehicles are selected without considering their completion probability. This reduces the completion probability of the vehicles, as shown in Fig. 5, leading to fewer participating vehicles, and resulting less trained data and higher loss values. DA achieves a better FL performance than Random because it selects vehicles based on the size and distribution of their collected data to optimize model performance. However, because it does not consider the reduced transmission rates during mobility, disconnections between vehicles and the RSU, or failed model-parameter transmissions, its FL performance is lower than that of ERA, MACA, and the proposed method. ERA achieves an improved loss value and test accuracy compared to MACA but remains lower than the proposed algorithm, as shown in Fig. 2. This shows that leveraging V2V in FL improves the accuracy of the global model by collecting additional updates from selected vehicles, but it also requires optimized V2V resource allocation to forward these updates to the RSU with minimal transmission delay.

The proposed algorithm also outperforms the benchmark methods for non-IID datasets from both the MNIST and FMNIST datasets, as shown in Fig. 3. Compared with Random, DA, MACA, and ERA, the proposed algorithm achieves average accuracy improvements of 16.1%, 17.3%, 15.0%, and

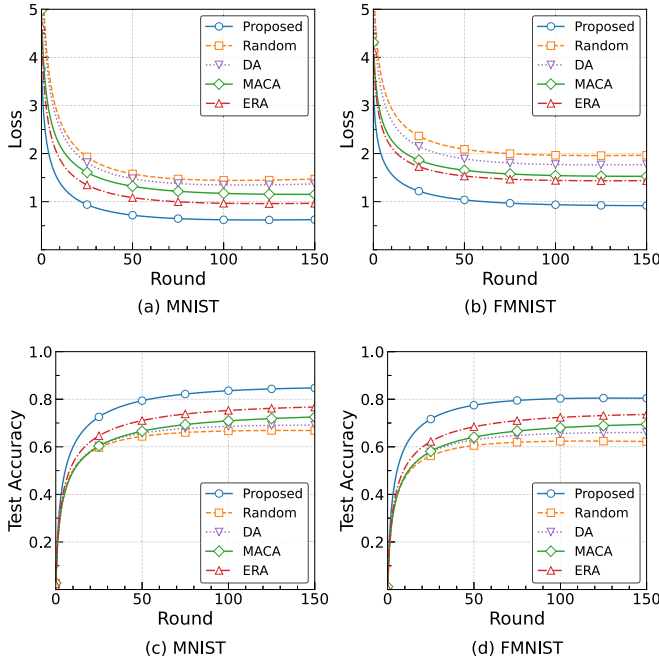


Fig. 3. FL performance comparison of different schemes over the MNIST and FMNIST datasets for non-IID.

11.4% on MNIST, and 20.7%, 19.3%, 16.5%, and 13.1% on FMNIST, respectively. Although the accuracy of all the algorithms decreases under non-IID data, the drop is smaller for Random and DA than for the other baselines because uniform random client sampling promotes greater participation diversity, while the data-aware selection in DA helps reduce training data imbalance. However, because Random and DA do not consider the completion probability in vehicle selection, more vehicles fail to transmit model parameters to the RSU, thereby reducing the FL performance relative to the proposed algorithm, MACA, and ERA. Although MACA considers vehicles' completion probability during selection and thus achieves better accuracy than Random and DA, it still has a lower performance than the proposed method and ERA because it does not use model updates from vehicles that left the coverage area of the RSU.

Fig. 4 illustrates the total delays required to complete the entire FL training process until the model converges at various CPU frequencies of the vehicle. The total delay decreases as the vehicle's CPU frequency increases because the time required for local training is reduced. The proposed algorithm achieves the lowest total delay compared to the other four methods, even when more vehicles participate in the FL process. The proposed algorithm achieves the lowest total delay, even with more participating vehicles. This is because the proposed algorithm not only optimizes transmission rates for transmitting the model parameter through resource allocation but also increases the completion probability, reducing straggler effects and the waiting time for the final updates at the RSU. Although ERA achieves better FL performance than MACA, as shown in Fig. 2 and Fig. 5, it incurs a higher total delay. This is because the random allocation of communication resources for V2V communication increases the time required

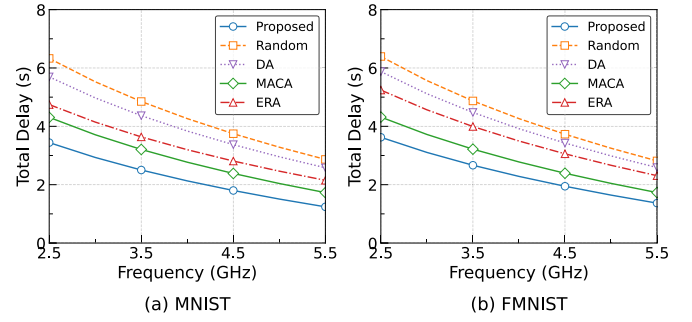


Fig. 4. Total delay comparison of different schemes over the (a) MNIST and (b) FMNIST datasets.

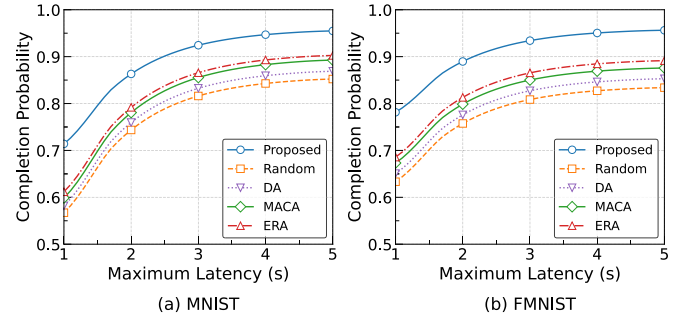


Fig. 5. Average completion probability comparison of different schemes over the (a) MNIST and (b) FMNIST datasets.

for the RSU to receive all the model parameters. Since MNIST and FMNIST have the same number of samples and we use the same CNN model, the total local training and transmission delays are similar, leading to similar completion probabilities, even though the datasets differ, as shown in Fig. 5.

As shown in Fig. 5, the completion probability increases with the maximum latency allowed for local training and the parameter forwarding increases for each method, whereas the proposed method consistently achieves the highest completion probability. The proposed method achieves a higher completion probability than the other methods because it selects vehicles whose completion probability exceeds a predefined threshold. In other words, it selects clients expected to complete local training and forward their updates, leveraging V2V relaying when a vehicle is likely to leave the coverage area of the RSU. Although the ERA utilizes V2V communication while MACA does not, the difference in completion probability is small because ERA uses the same radio resources, even though the distances between vehicles are different, leading to long transmission times for some transmissions of the updated model. In contrast, the proposed algorithm optimizes the communication resources based on the velocity, location, and expected sojourn time of the vehicle as well as the distances from other vehicles to minimize the total delay taken for one round of FL, leading to the highest completion probability compared to other algorithms. A lower completion probability implies that more updates fail to reach the RSU and are therefore excluded from global aggregation, leading to wasted computing resources and reduced model performance, as shown in Figs. 2 and 3.

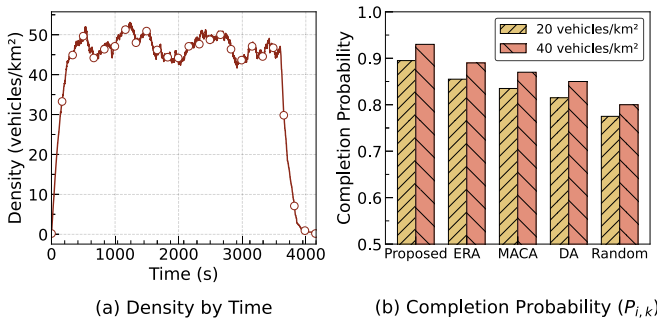


Fig. 6. (a) Vehicle density over time. (b) Completion probability at vehicle densities of 20 and 40 vehicles/km².

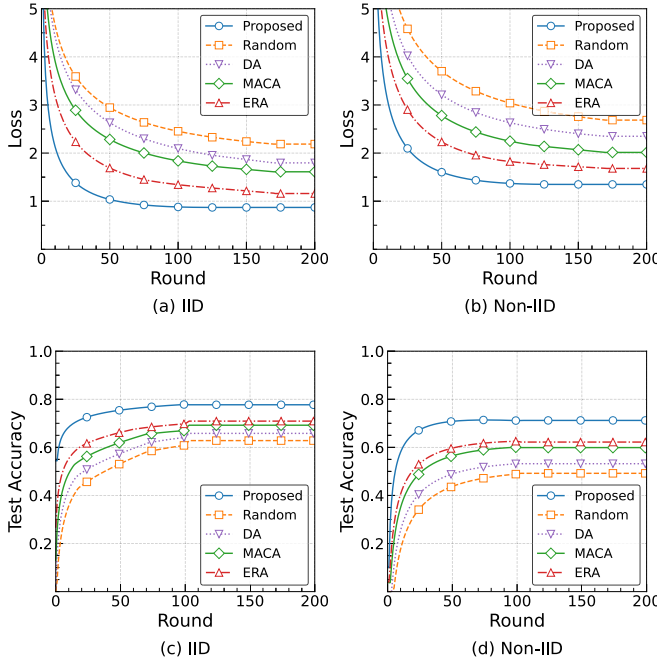


Fig. 7. FL performance comparison of different schemes over the LISA traffic light datasets for IID and non-IID.

Fig. 6 shows the change in vehicle density over time during the simulation, along with completion probability for different densities. As illustrated in Fig. 6 (a), the vehicle density remains between 40 and 50 vehicles/km², but at the beginning and end of the simulation it drops sharply, resulting in an average density of around 20 vehicles/km². Although the proposed algorithm achieves a higher completion probability than the baselines across vehicle densities (Fig. 6 (b)), a sudden drop to 20 vehicles/km² reduces the completion probability for all methods compared with 40 vehicles/km². This occurs because fewer vehicles lead to an average velocity increase of approximately 15–20%, thereby reducing the sojourn time within the coverage area of the RSU. In addition, a sudden decrease in vehicle density reduces the number of suitable vehicles for forwarding through V2V communication, which limits the improvement in the completion probability that can be achieved through multi-hop V2V transmission.

2) *Evaluation on LISA Traffic Light Dataset*: Fig. 7 shows the loss and accuracy values of the global model on the LISA

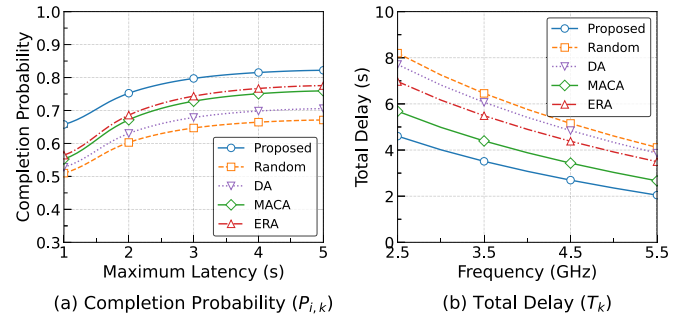


Fig. 8. Average completion probability and total delay comparison of different schemes over the LISA traffic light datasets.

traffic light datasets. Compared with MNIST and FMNIST, all algorithms show higher global loss values and lower test accuracy on LISA. This is because the traffic lights in LISA are small and off-center in high-resolution images (e.g., 1280 × 960), whereas MNIST and FMNIST consist of centered 28 × 28 crops, which makes classification with a lightweight CNN more challenging. In the simulations, the proposed method achieves the highest test accuracy under both IID and non-IID conditions compared with the other algorithms because more vehicles successfully transmit updated local parameters, allowing the global model to adapt more effectively to heterogeneous data. Specifically, the proposed algorithm outperforms ERA, MACA, DA, and Random by 8.5%, 11.5%, 15.6%, and 19.4%, respectively, on the IID dataset, and by 10.4%, 13.3%, 20.5%, and 25.1% on the non-IID dataset, respectively. In LISA, the FL performance gaps between the proposed algorithm and benchmarks are larger under the non-IID split because the dataset contains more classes with greater variation in viewpoints, scales, and backgrounds compared to MNIST/FMNIST. Therefore, models trained on limited, skewed data are more likely to perform poorly on classes they have seen less frequently during training, which reduces overall accuracy.

Fig. 8 shows the average completion probability and total delay for FL under different maximum aggregation latency values and vehicle CPU frequencies, respectively. As shown in Fig. 8 (a), the completion probability for LISA is lower than for MNIST/FMNIST because training on the higher-resolution LISA images requires more computation time, which reduces the number of vehicles that can finish local training and transmit their updated model parameters while within the RSU's coverage. Consequently, the total delay and completion probability of Random and DA degrade significantly on LISA compared to MNIST/FMNIST, as the heavier computational load causes more vehicles to leave the coverage area of the RSU before transmitting their updates when completion probability is not considered. While the completion probability is lower on LISA than on MNIST/FMNIST for every method, the proposed algorithm achieves the highest completion probability and the lowest total delay by not only selecting vehicles likely to successfully complete local training and forward their model parameters, but also by optimizing communication resources to further improve the completion probability of vehicles.

The simulation results show that the V2V-FL mechanism can improve the accuracy of the global model by increasing the completion probability through V2V communications. ML-based smart services are increasingly applied not only to autonomous driving and smart city applications such as traffic prediction, route planning, and surveillance but also to vehicle resource optimization, including the allocation of communication and energy resources. Hence, the V2V-FL mechanism is expected to enhance service quality and reliability in these applications.

VII. CONCLUSION

In this paper, we propose a V2V-FL mechanism that leverages multi-hop V2V communication to enhance the completion probability of vehicles. To evaluate whether a vehicle can successfully forward the updated model to the RSU either directly or via multi-hop V2V communication, we model its completion probability while considering mobility, as well as communication and computing resources. To minimize the total delay and maximize the completion probability, we formulate an updated model forwarding decision and resource allocation problem. Since the proposed problem is a mixed-integer non-convex optimization problem, we solve it through decomposition using KKT conditions and a heuristic method. Simulation results on both baseline datasets (MNIST and FMNIST) and the vehicular dataset (LISA traffic light) confirm that the proposed V2V-FL scheme outperforms the benchmark algorithms by achieving a lower total delay and a higher completion probability.

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